

BrightLearn Data Agent Assignment

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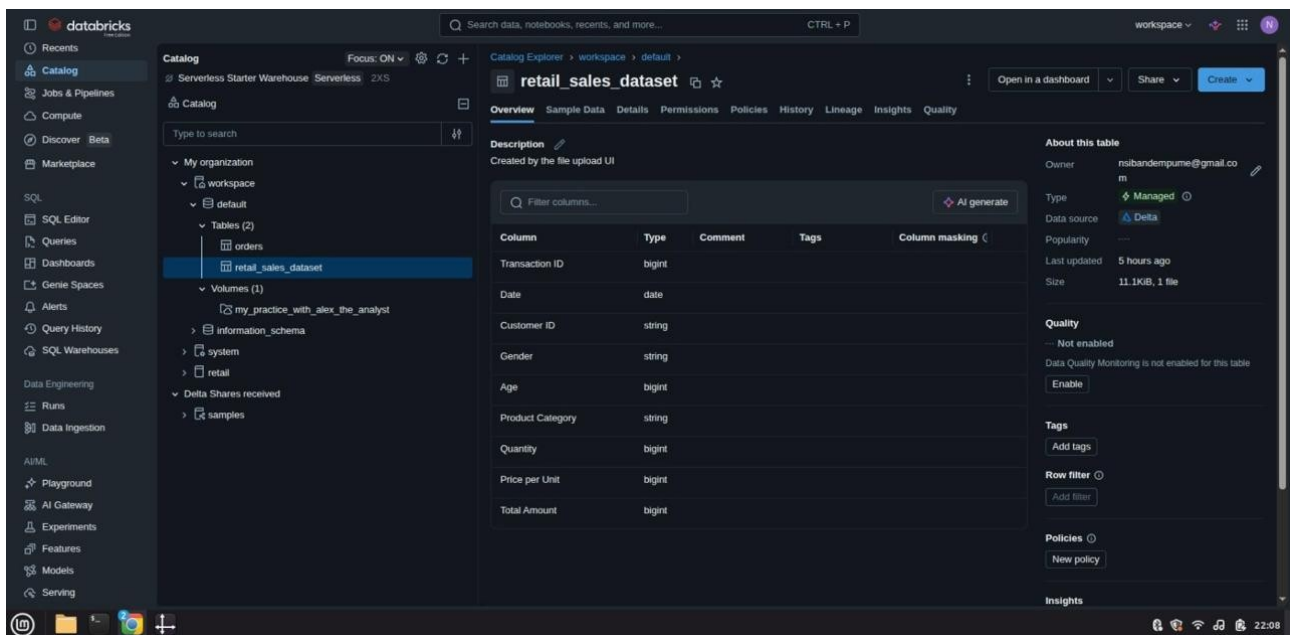
Introduction & Objective

This is a Data Agent building project that I have build after learning on data agent with Bright Learn. The objective was to apply the knowledge and skills gained in class by designing and building a data agent from scratch using Databricks.

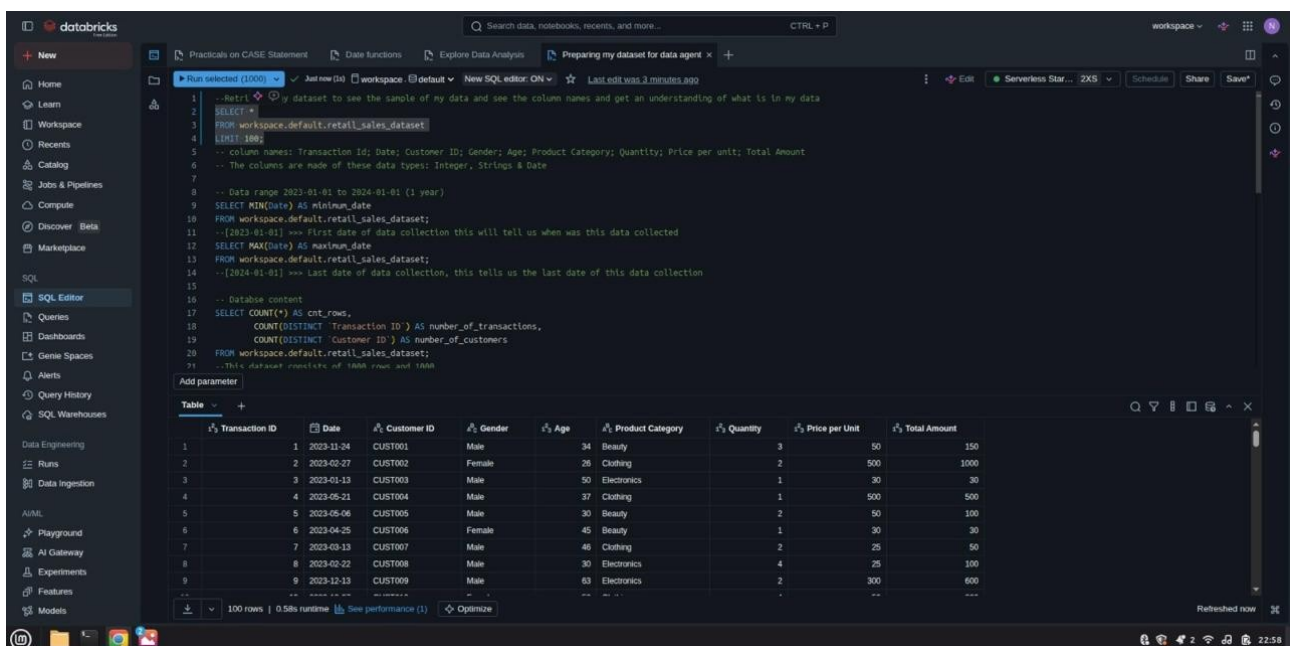
My Data Agent is called sales Intelligence Agent. This agent is designed to explore and analyse retail sales data for a store, helping the store owner, managers, and decision makers answer key business questions about sales performance, product categories, and customer purchasing behaviour. Through this project I was able to practise writing SQL queries, data visualization, validating data, and interpreting results to support real business decisions. Data visualization was a new skill I was introduced to during this project and I look forward to developing it further.

Dataset description

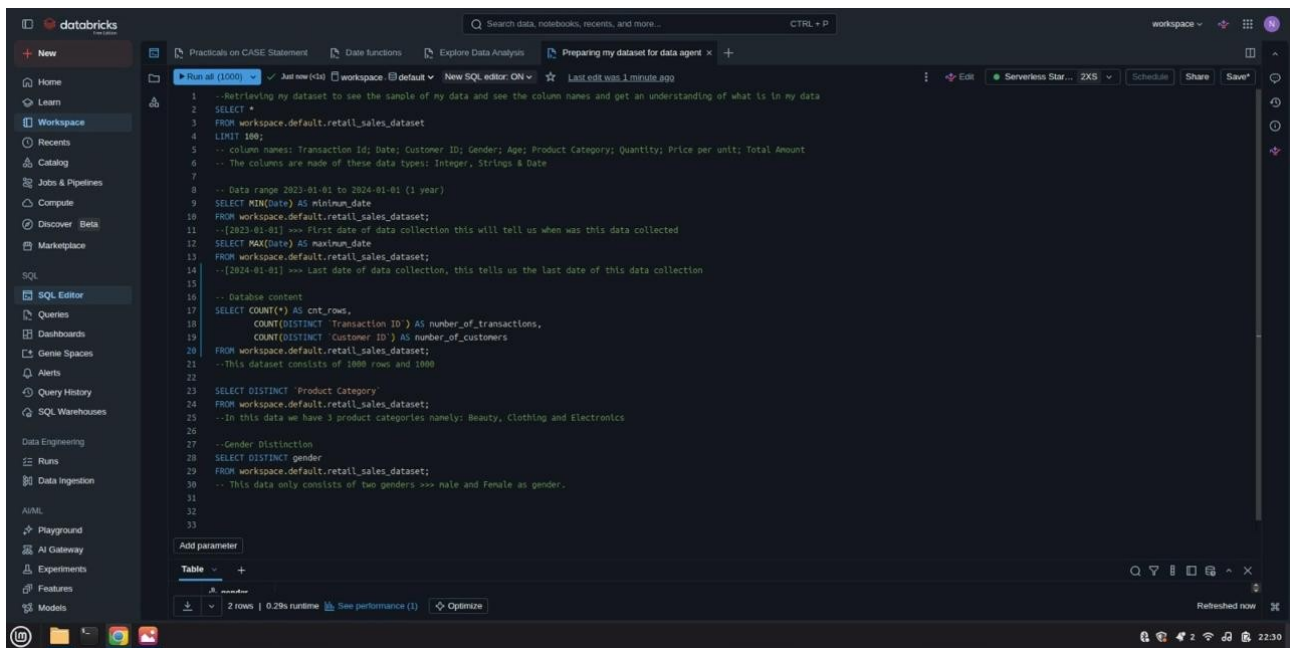
The dataset I used was provided to us as Bright Learn students and it contains one thousand Columns and rows. This dataset shows transactions collected between January 2023 and January 2024, covering information such as product categories, customer demographics, transaction amounts, and purchase dates.



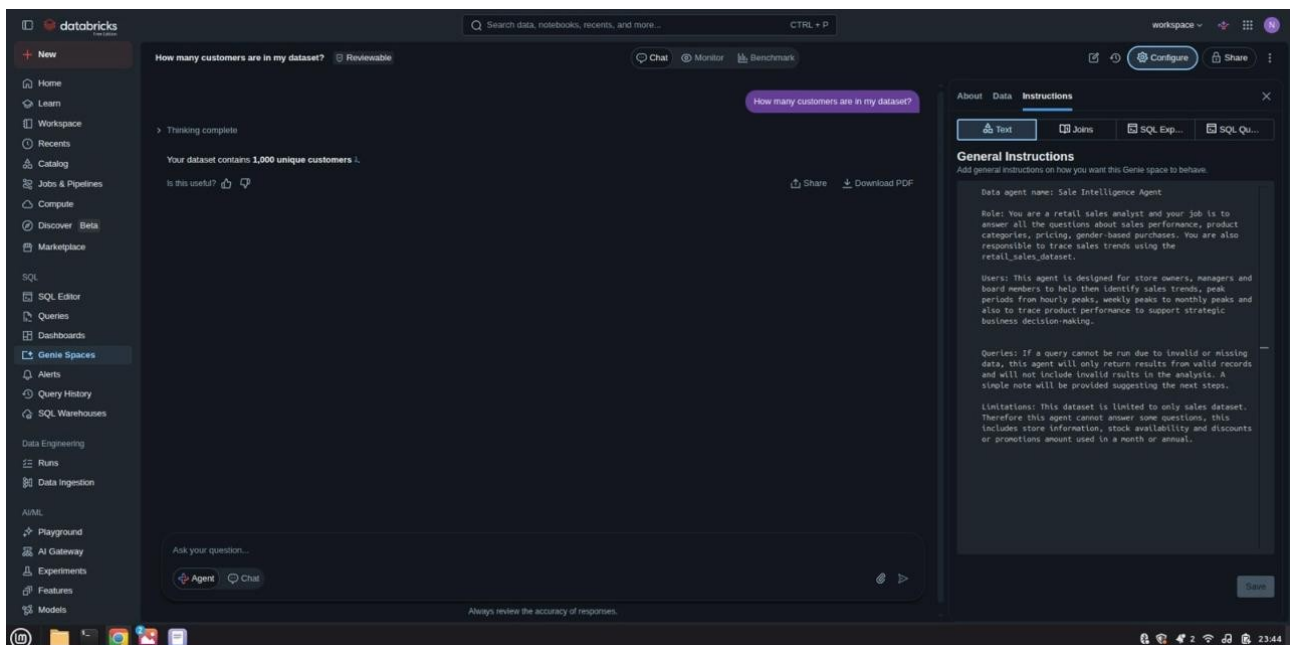
Retail Sales Dataset successfully uploaded on databrick. Screenshot



Retail Sales Dataset table prep screenshot.



Dataset review screenshot



Data Agent instructions screenshot

Data Agent Instructions: Sales Intelligence Agent

Role: You are a retail sales analyst and your job is to answer all the questions about sales performance, product categories, pricing, and gender-based purchases. You are also responsible for tracing sales trends using the retail_sales_dataset.

Users: This agent is designed for store owners, managers, and board members to help them identify sales trends, peak periods from hourly peaks, weekly peaks to monthly peaks, and also to trace product performance to support strategic business decision-making.

Queries: If a query cannot be run due to invalid or missing data, this agent will only return results from valid records and will not include invalid results in the analysis. A simple note will be provided suggesting the next steps.

Limitations: This dataset is limited to only the sales dataset. Therefore this agent cannot answer some questions, this includes store information, stock availability, and discounts or promotions amount used in a month or annually.

12 Common questions

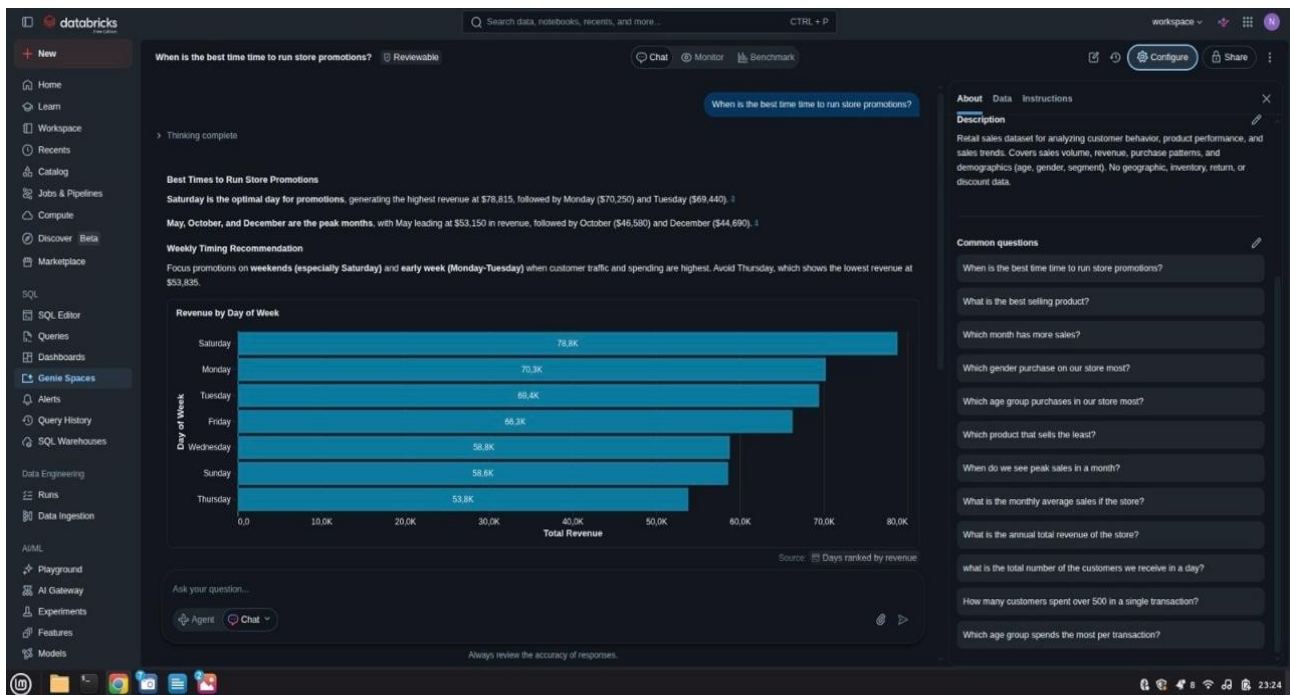
1. When is the best time to run store promotions?
2. What is the best selling product?
3. Which month has more sales?
4. Which gender purchases in our store most?
5. Which age group purchases in our store most?
6. Which product sells the least?
7. When do we see peak sales in a month?
8. What is the monthly average sales of the store?
9. What is the annual total revenue of the store?
10. What is the total number of customers we receive in a day?
11. How many customers spent over 500 in a single transaction?
12. Which age group spends the most per transaction?

Data Agent Answer Validation

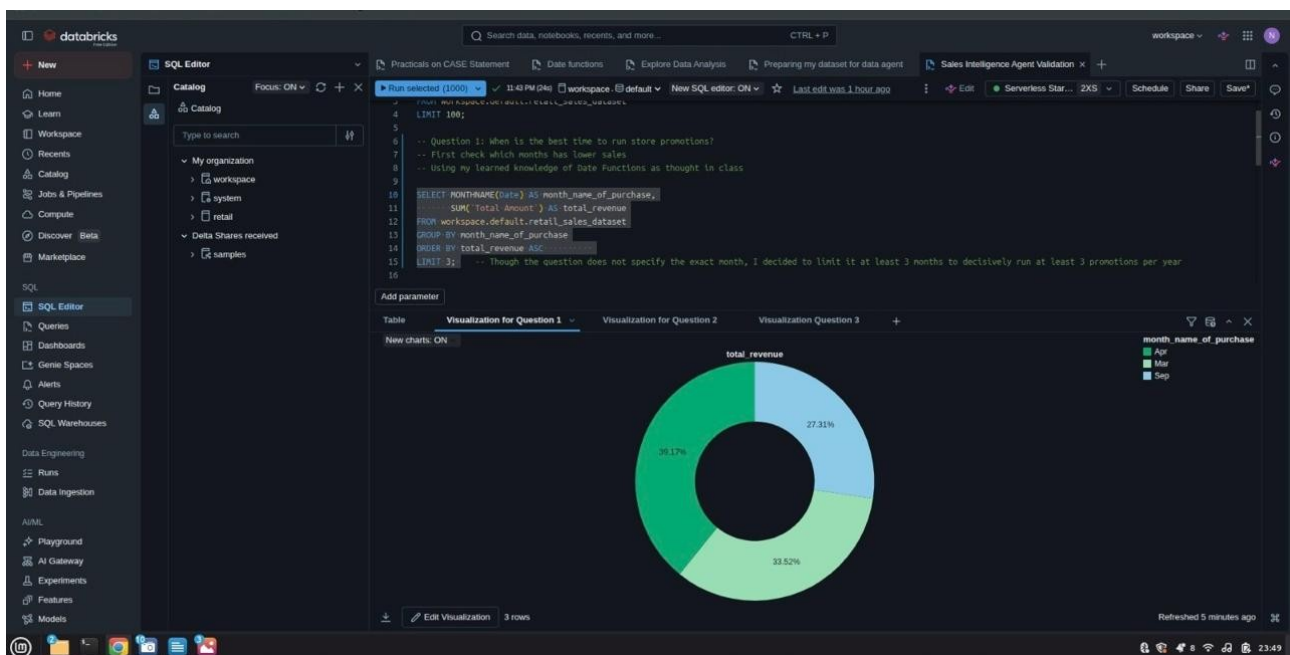
To validate the answers given by the Sales Intelligence Agent, I wrote SQL queries for each question. Then I created visualizations from the query results and compared them against what the agent returned.

Validation to question 1: When is the best time to run store promotions?

The agent looked at the month with less sales as what the store can use to bring up sales, so September was the weakest month according to the data agent. On my query I also looked for months that brings less sales and I ran a query which returned the 3 lowest revenue months and the bar chart confirmed September had the shortest bar, meaning the lowest sales.



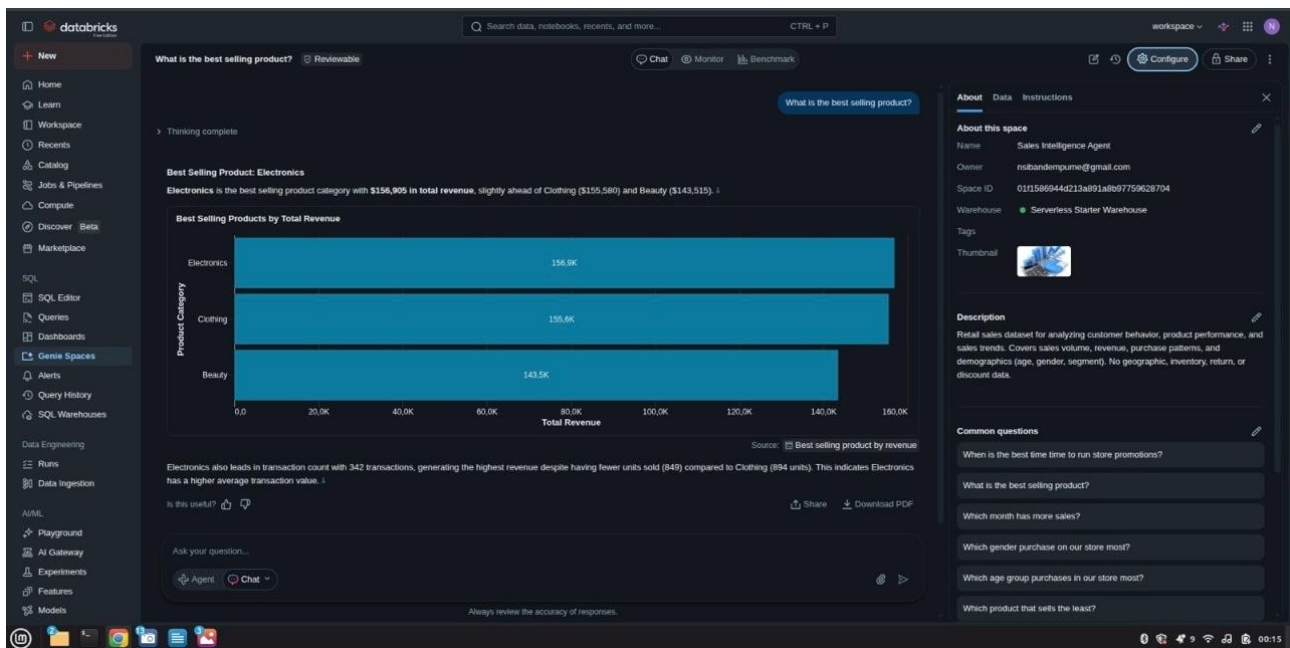
Data Agent response to question 1 screenshot.



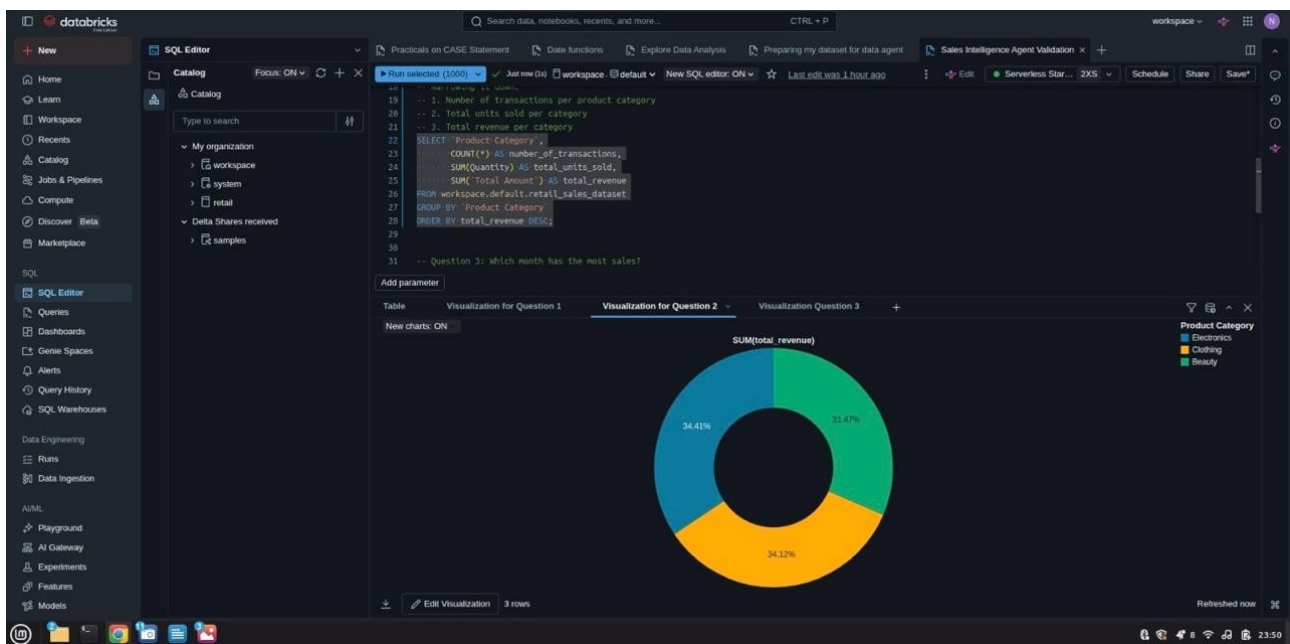
Data Agent answer to question 1 validation.

Validation to question 2: What is the best selling product?

The agent said Electronics was the best selling category at \$156,905 in revenue. My query grouped sales by product category and ordered by total revenue. The pie chart confirmed Electronics at 34.41%, sitting just ahead of Clothing at 34.12% and Beauty at 31.47%.



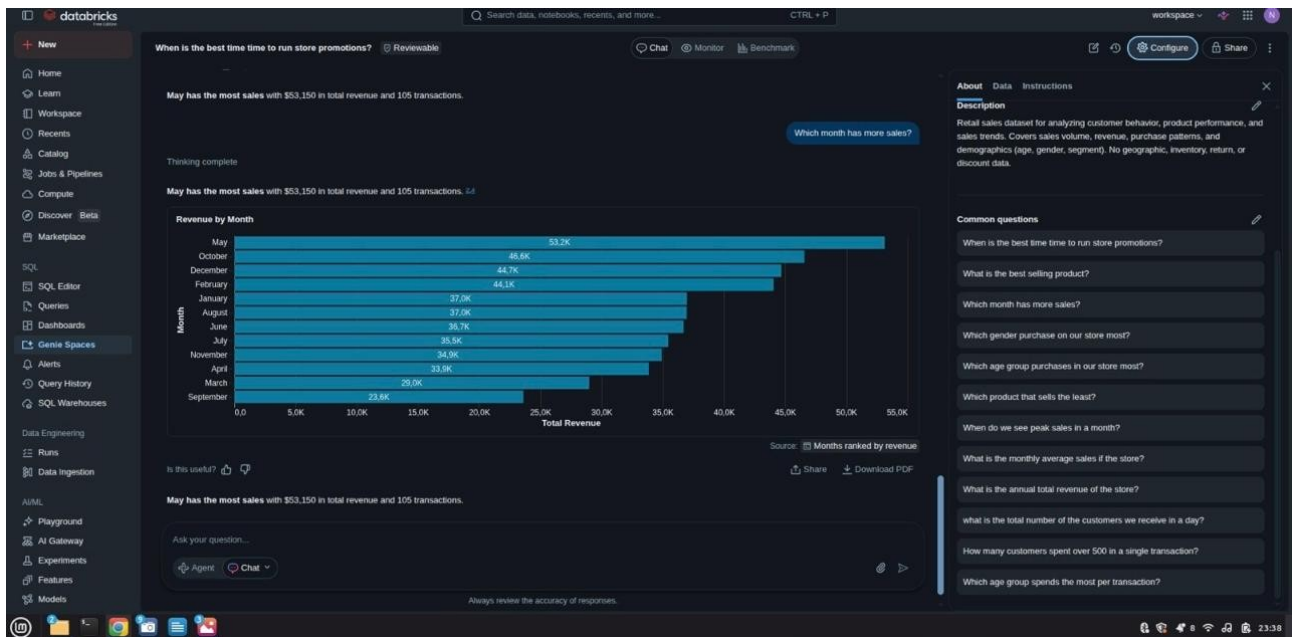
Data Agent response to question 2 screenshot.



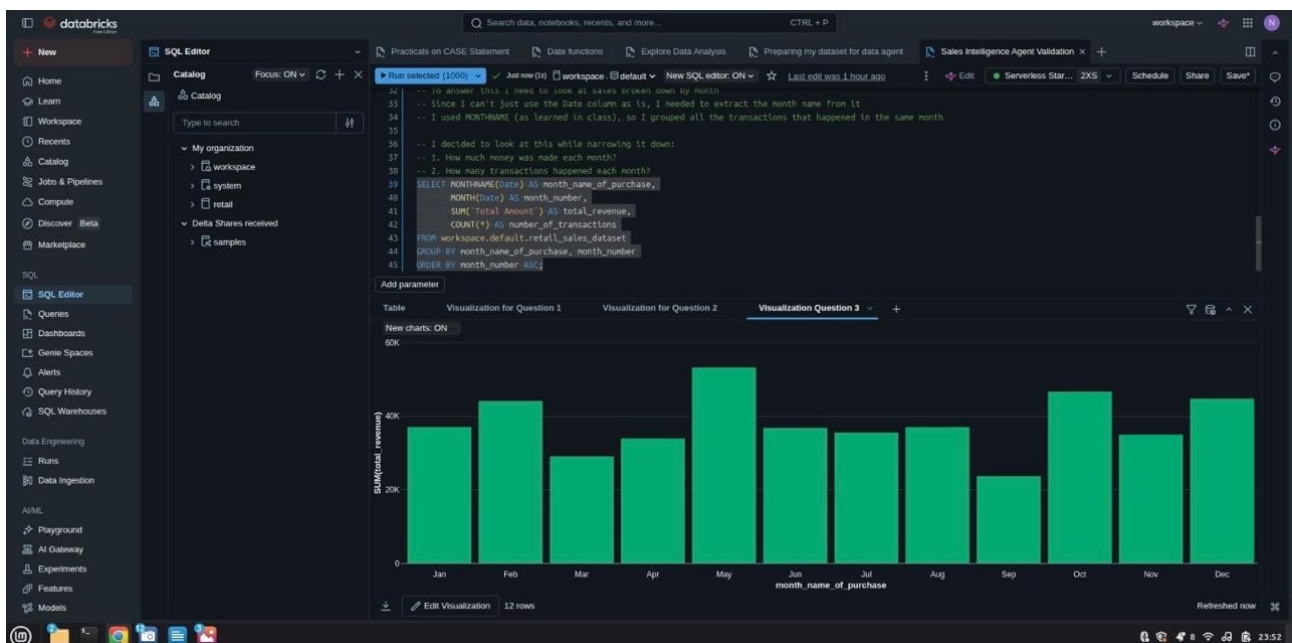
Data Agent answer to question 2 validation.

Validation 3: Which month has the most sales?

The agent said May was the strongest month at \$53,150. On my query I also extracted the month name from the date column and grouped transactions by month. The bar chart clearly showed May as the tallest bar and September as the shortest, matching the agent exactly.



Data Agent response to question 3 screenshot.



Data Agent answer to question 3 validation.

Key Insights

After asking my Data Agent all 10 questions, a few important patterns started to stand out about how the store is performing.

The first thing I noticed came from comparing the best selling and lowest selling product categories. Electronics brought in the highest revenue at \$156,905, with Clothing very close behind at \$155,580, while Beauty generated \$143,515. What stood out to me was how small the difference actually is between the highest and lowest category. A gap of only \$13,390 over an entire year shows that the store has a fairly balanced product range instead of depending heavily on one category. At the same time, Beauty clearly needs more attention because it records the lowest

number of sales and transactions, even though customers tend to spend more per purchase there than they do on Clothing.

The second insight became clear when comparing the strongest sales month with the weakest one. May performed exceptionally well with \$53,150 in sales, while September dropped to only \$23,620, which is less than half of May's revenue. This tells me the store's sales performance changes quite dramatically throughout the year. It is not just a small seasonal difference but a major shift that could affect planning, staffing, stock levels, and promotions if the business is not prepared for it.

The third insight came from looking at customer age groups and spending habits. Customers aged 45–54 contribute the most overall revenue and transactions, making them the store's most reliable customer group. However, the younger customers under 25 revealed a very interesting trend. Even though they visit the store the least, with only 149 transactions, they spend the most money per visit at an average of \$501 per transaction. This suggests that younger shoppers may buy less frequently, but when they do shop, they are willing to spend significantly more, creating a strong opportunity for targeted marketing and promotions aimed at that age group.

Business Recommendation

I would recommend the store to consider running promotions around September, March, and April since these are the weakest months. Running targeted campaigns during these times such as discounts on Electronics which drops the most during slow months, would help bring in customers who might otherwise not visit the store at all. Thursdays are also the weakest day of the week, so a mid-week special or loyalty reward on Thursdays could help drive foot traffic on that day specifically.

Since the under-25 age group has the highest average spend per transaction but the lowest number of visits, the store should invest in marketing that targets younger customers. Things like social media campaigns or student promotions would encourage the youngsters to visit the store more often, and given how much they spend when they do come in, this could meaningfully increase revenue.

Beauty is the least performing category not because customers spend less per visit, but because fewer customers are buying it. The store could look at bundling Beauty products with Clothing or running Beauty-specific promotions to increase transaction volume in that category.

The late month surge, especially around the 23rd to 24th, tells me customers are shopping after pay day. The store should make sure shelves are well stocked and staff levels are higher during the last 10 days of each month to make the most of the month end peak.

Finally, since Saturday is the strongest day of the week at \$78,815 in revenue, the store should make sure it is fully prepared every Saturday with enough staff, stock, and an inviting in-store experience to maximise what is already a high-traffic day.

Conclusion

For me this project was a great learning experience and I feel like I came out of it with skills I did not have going in. I remember my first time trying to build a data agent at an event sponsored by Google. I had no idea what was happening, and the part that confused me the most was how agents actually connect to data. It felt like a concept that was completely out of reach for me at the time. Bright Learn changed that. By the time I was building the Sales Intelligence Agent on Databricks, I understood exactly how the agent connects to the dataset, reads it, and returns meaningful answers from it. That shift in understanding is one of the things I will forever cherish from this programme.

The validation process was also something I found very valuable, challenging at first but eventually I got the hang of it. Going back to write SQL queries myself and comparing them against the agent's answers gave me confidence that the results were accurate and not just taken at face value.

What was also hard for me was data visualisation. This was a brand new skill I was introduced to briefly in class, so during this project I found it challenging at first. Having to decide which chart type to use that will best represented each answer was kind of a challenge.

If I could change something, I would ask more specific questions from the beginning. Some of my questions were quite broad and the agent gave very detailed answers that I had to work through carefully. Next time I would break broader questions into smaller, more focused ones so that the answers are easier to validate and act on.

This project showed me that a well-built data agent can genuinely help a business make smarter decisions.